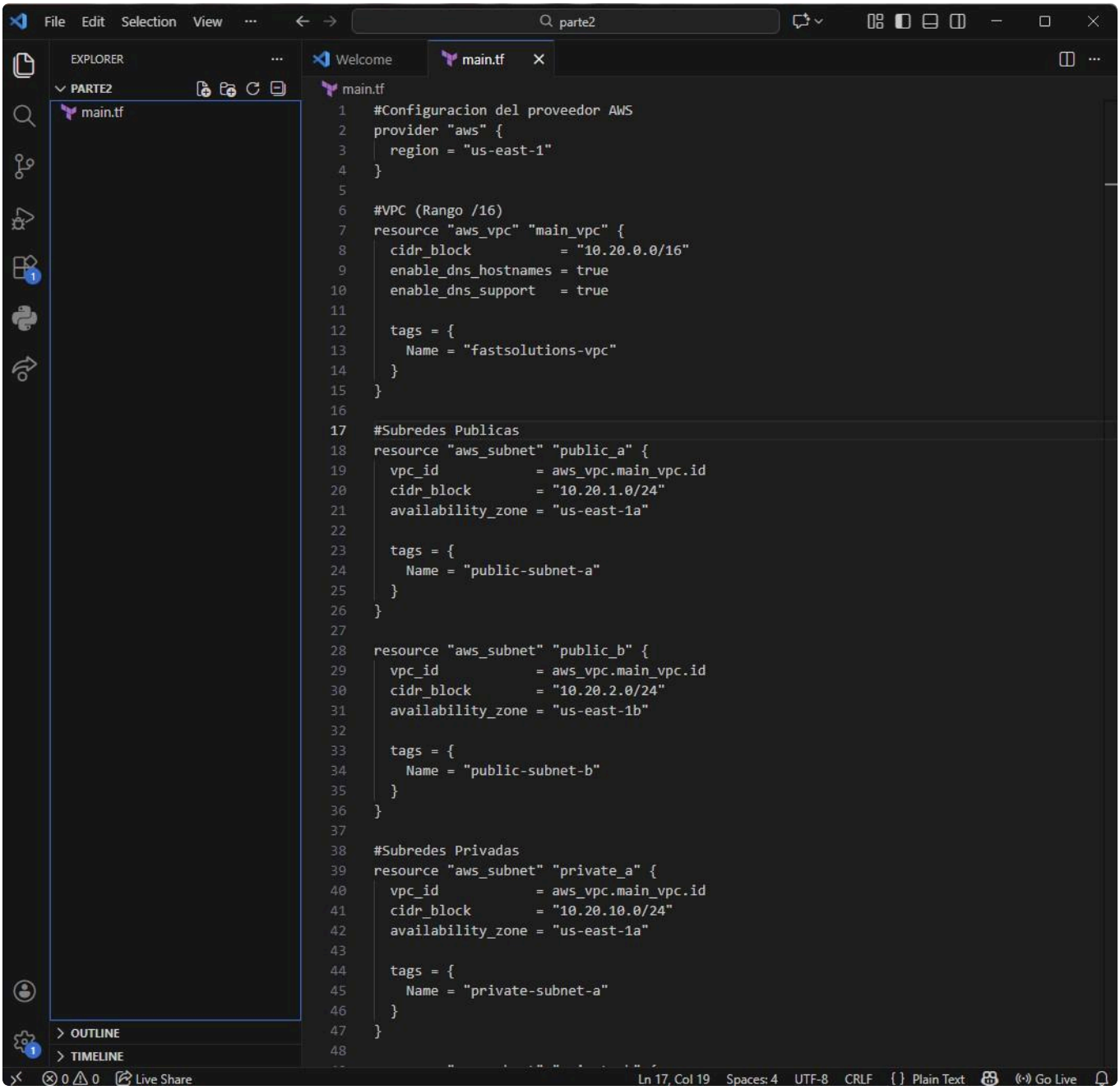
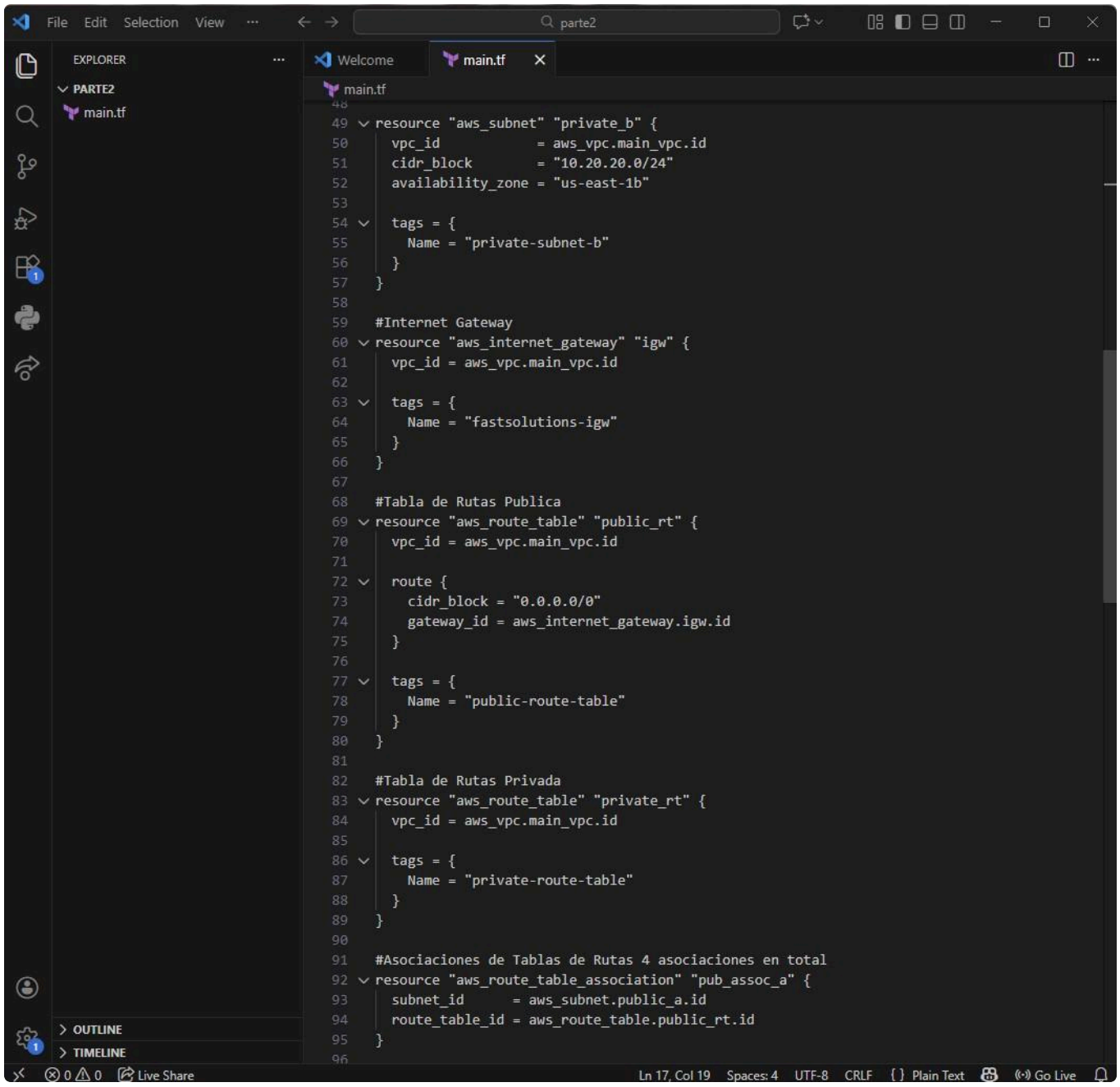
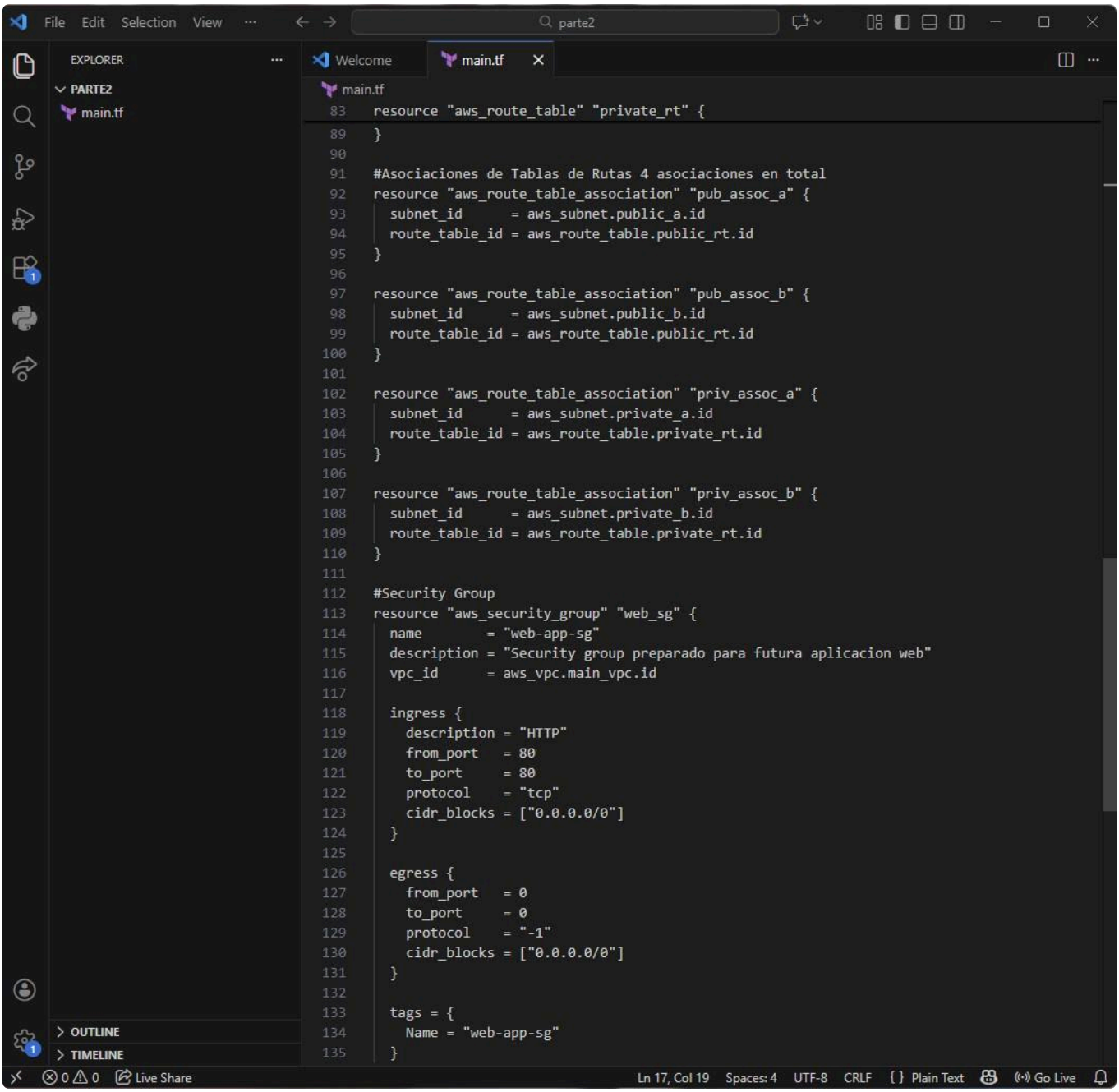
# fastsolutions-ha-network

Este documento presenta la implementación práctica de una arquitectura de red en AWS para el caso de estudio de FastSolutions SpA, desarrollada íntegramente mediante Infrastructure as Code (IaC) con Terraform. Incluye el diseño técnico, el proceso de despliegue automatizado validado mediante la terminal y la verificación de los recursos de red y seguridad en la consola de AWS.

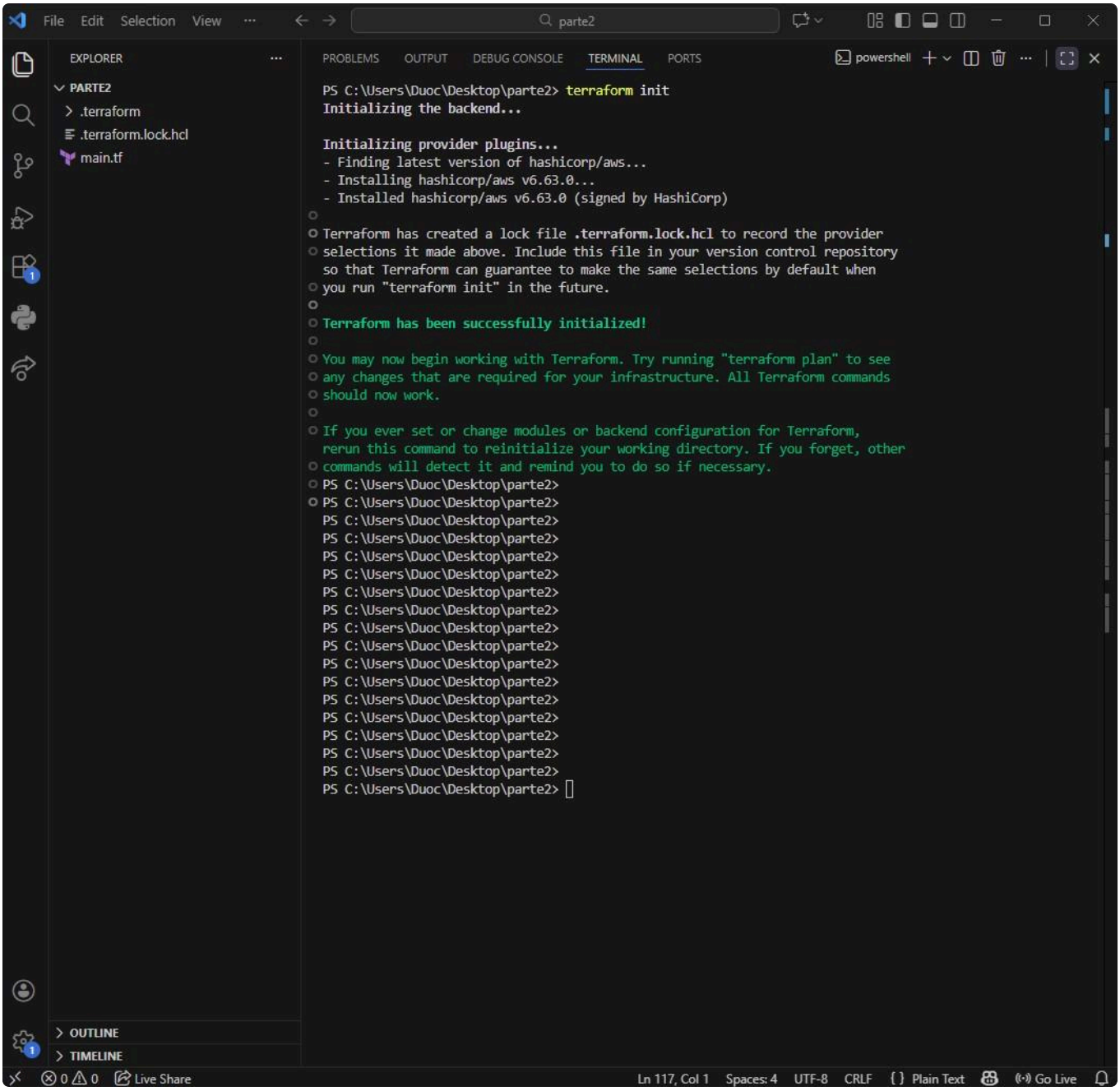
Código de Terraform

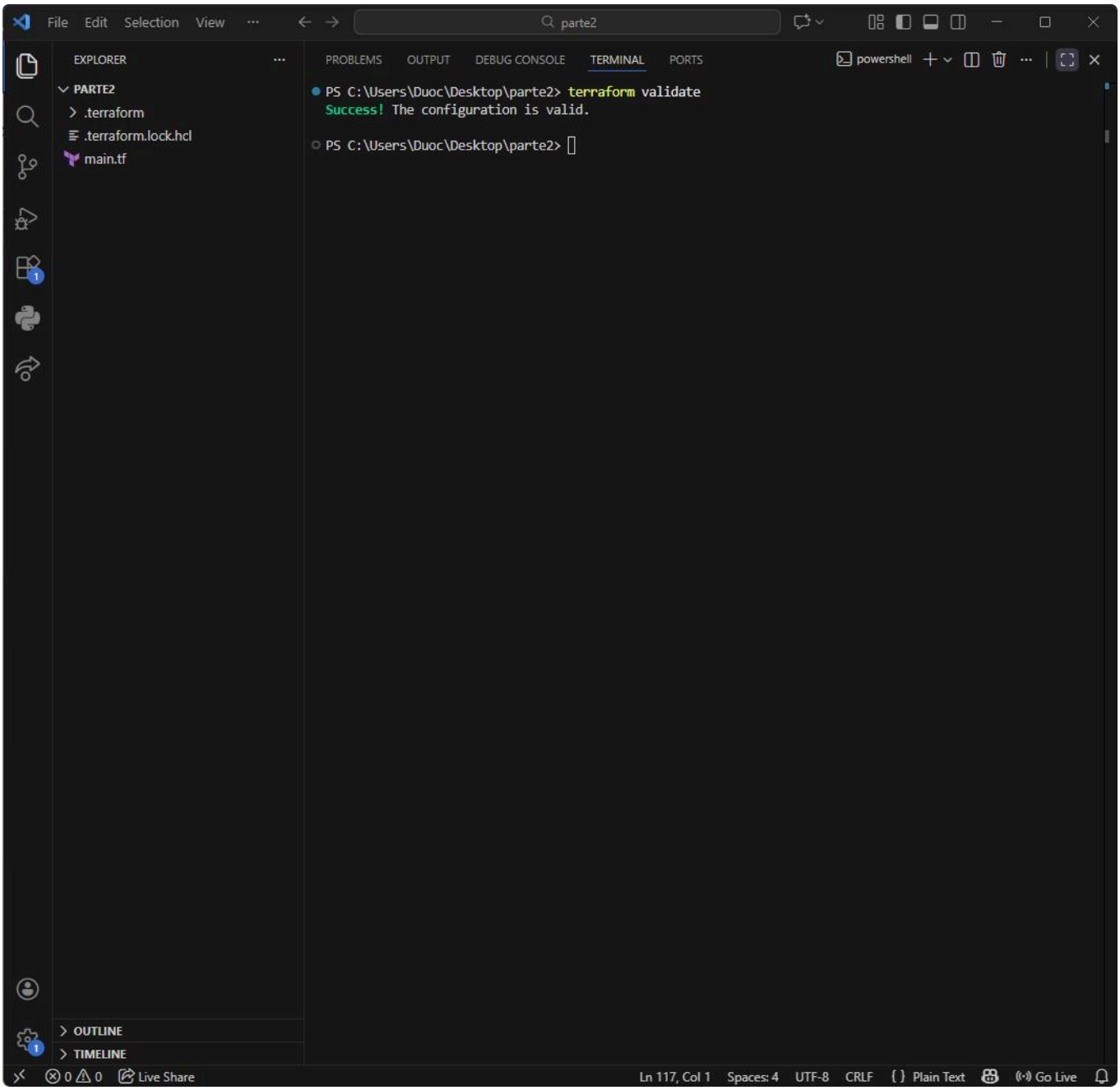


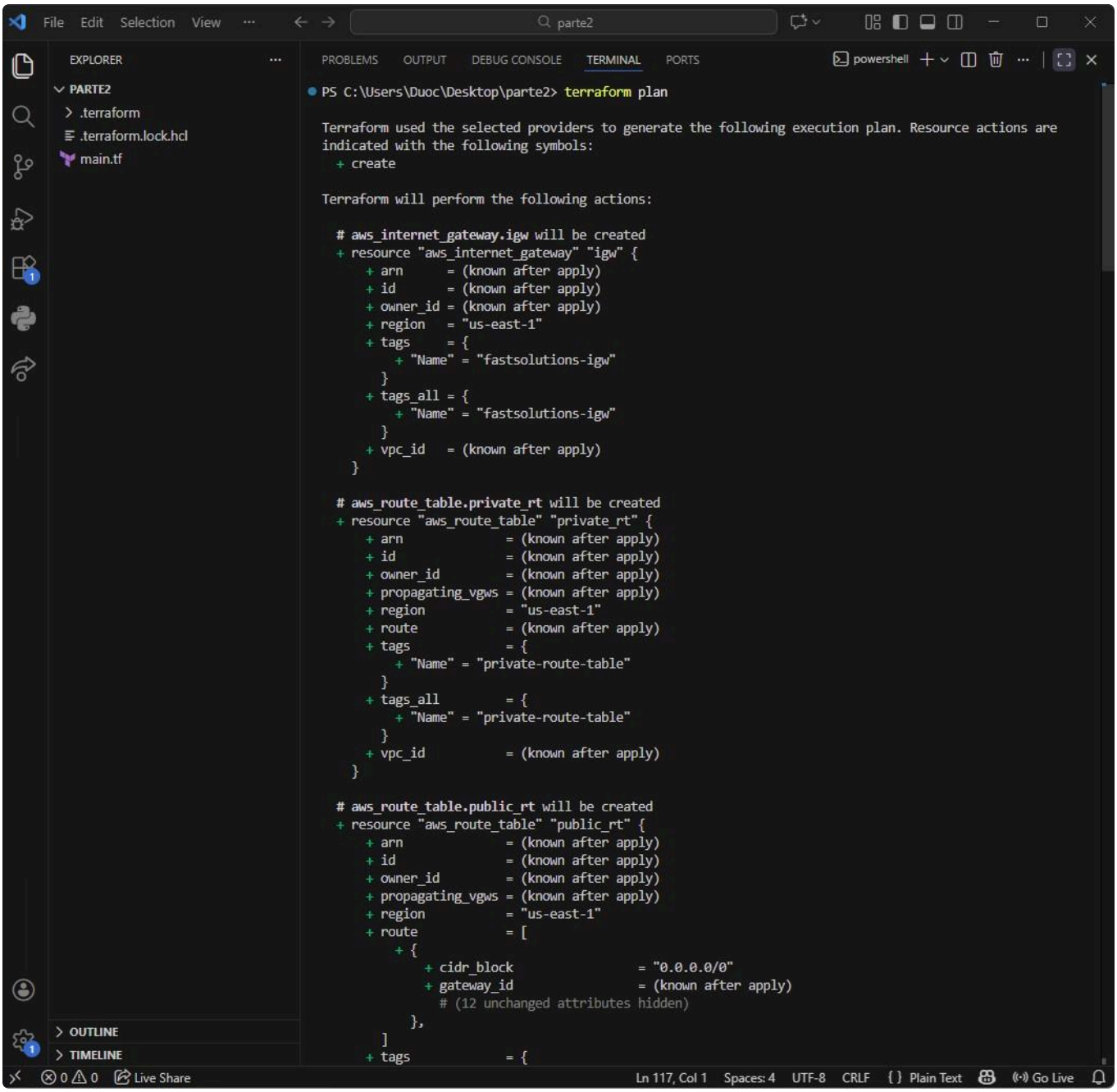

File Edit Selection View
parte2
EXPLORER
PARTE2
main.tf
Welcome
main.tf
main.tf
resource "aws_subnet" "private_b" {
vpc_id = aws_vpc.main_vpc.id
cidr_block = "10.20.20.0/24"
availability_zone = "us-east-1b"
tags = {
Name = "private-subnet-b"
}
}
#Internet Gateway
resource "aws_internet_gateway" "igw" {
vpc_id = aws_vpc.main_vpc.id
tags = {
Name = "fastsolutions-igw"
}
}
#Tabla de Rutas Publica
resource "aws_route_table" "public_rt" {
vpc_id = aws_vpc.main_vpc.id
route {
cidr_block = "0.0.0.0/0"
gateway_id = aws_internet_gateway.igw.id
}
tags = {
Name = "public-route-table"
}
}
#Tabla de Rutas Privada
resource "aws_route_table" "private_rt" {
vpc_id = aws_vpc.main_vpc.id
tags = {
Name = "private-route-table"
}
}
#Asociaciones de Tablas de Rutas 4 asociaciones en total
resource "aws_route_table_association" "pub_assoc_a" {
subnet_id = aws_subnet.public_a.id
route_table_id = aws_route_table.public_rt.id
}
OUTLINE
TIMELINE
Live Share
Ln 17, Col 19 Spaces: 4 UTF-8 CRLF Plain Text Go Live

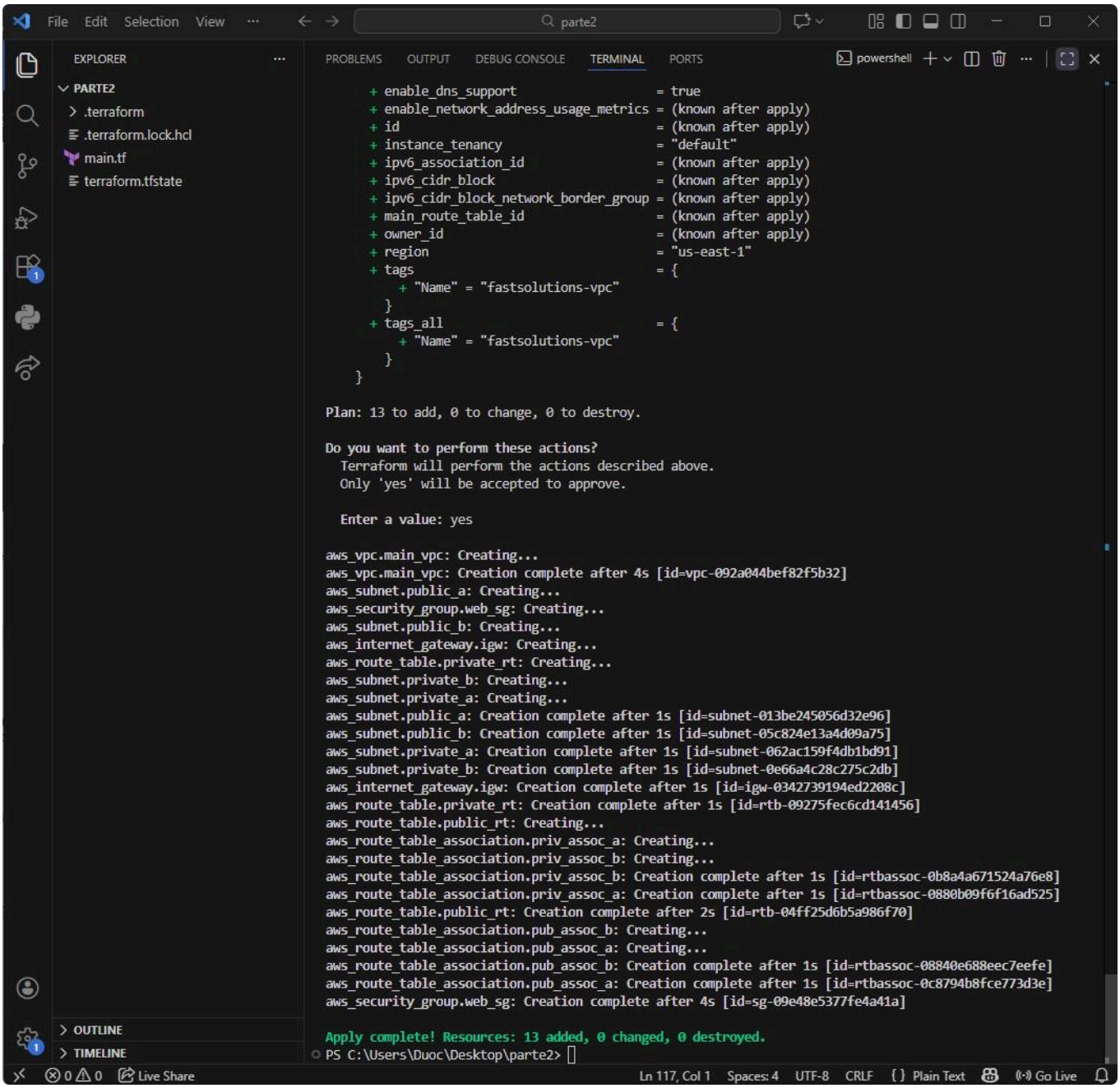

File Edit Selection View
parte2
EXPLORER
PARTE2
main.tf
Welcome
main.tf
main.tf
83 resource "aws_route_table" "private_rt" {
89 }
91 #Asociaciones de Tablas de Rutas 4 asociaciones en total
92 resource "aws_route_table_association" "pub_assoc_a" {
93 subnet_id = aws_subnet.public_a.id
94 route_table_id = aws_route_table.public_rt.id
95 }
97 resource "aws_route_table_association" "pub_assoc_b" {
98 subnet_id = aws_subnet.public_b.id
99 route_table_id = aws_route_table.public_rt.id
100 }
102 resource "aws_route_table_association" "priv_assoc_a" {
103 subnet_id = aws_subnet.private_a.id
104 route_table_id = aws_route_table.private_rt.id
105 }
107 resource "aws_route_table_association" "priv_assoc_b" {
108 subnet_id = aws_subnet.private_b.id
109 route_table_id = aws_route_table.private_rt.id
110 }
112 #Security Group
113 resource "aws_security_group" "web_sg" {
114 name = "web-app-sg"
115 description = "Security group preparado para futura aplicacion web"
116 vpc_id = aws_vpc.main_vpc.id
118 ingress {
119 description = "HTTP"
120 from_port = 80
121 to_port = 80
122 protocol = "tcp"
123 cidr_blocks = ["0.0.0.0/0"]
124 }
126 egress {
127 from_port = 0
128 to_port = 0
129 protocol = "-1"
130 cidr_blocks = ["0.0.0.0/0"]
131 }
133 tags = {
134 Name = "web-app-sg"
135 }
OUTLINE
TIMELINE
0 0 Live Share
Ln 17, Col 19 Spaces: 4 UTF-8 CRLF { } Plain Text Go Live

Terraform Plan / Apply

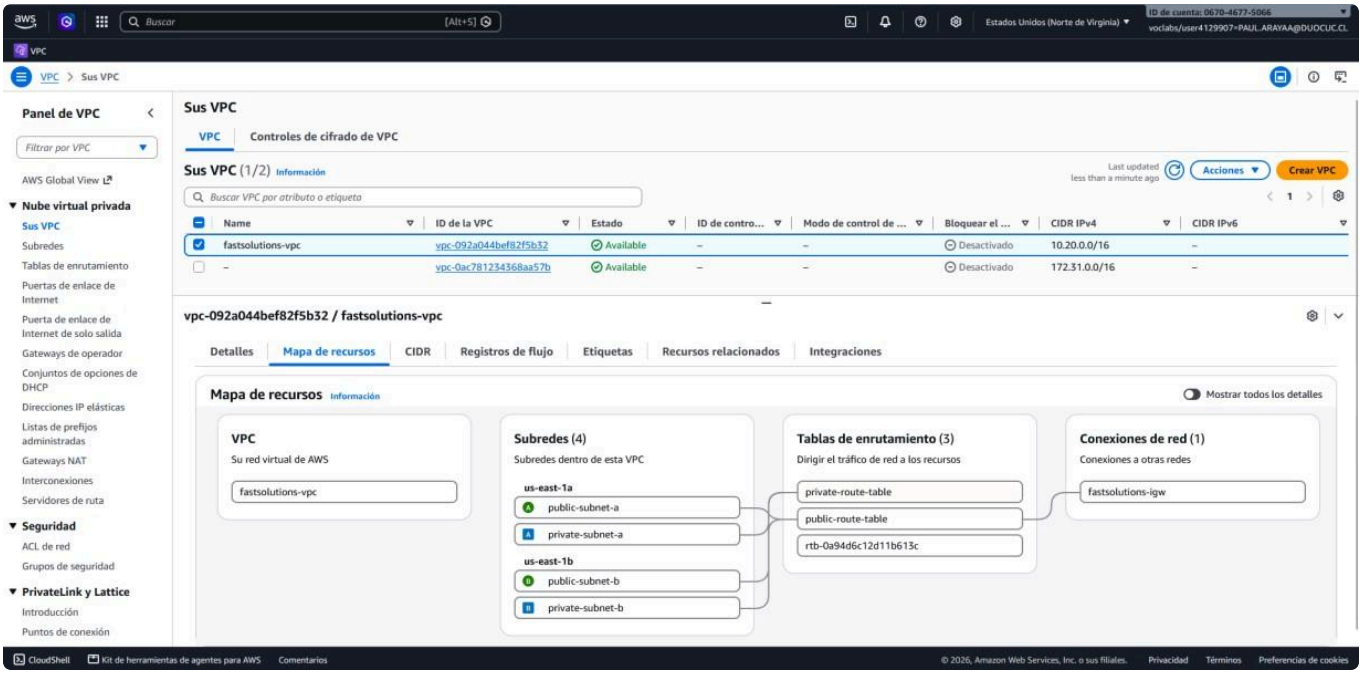



File Edit Selection View
parte2
EXPLORER
PARTE2
.terraform
.terraform.lock.hcl
main.tf
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
PS C:\Users\Duoc\Desktop\parte2> terraform validate
Success! The configuration is valid.
PS C:\Users\Duoc\Desktop\parte2>
OUTLINE
TIMELINE
0 0 Live Share
Ln 117, Col 1 Spaces: 4 UTF-8 CRLF { } Plain Text Go Live

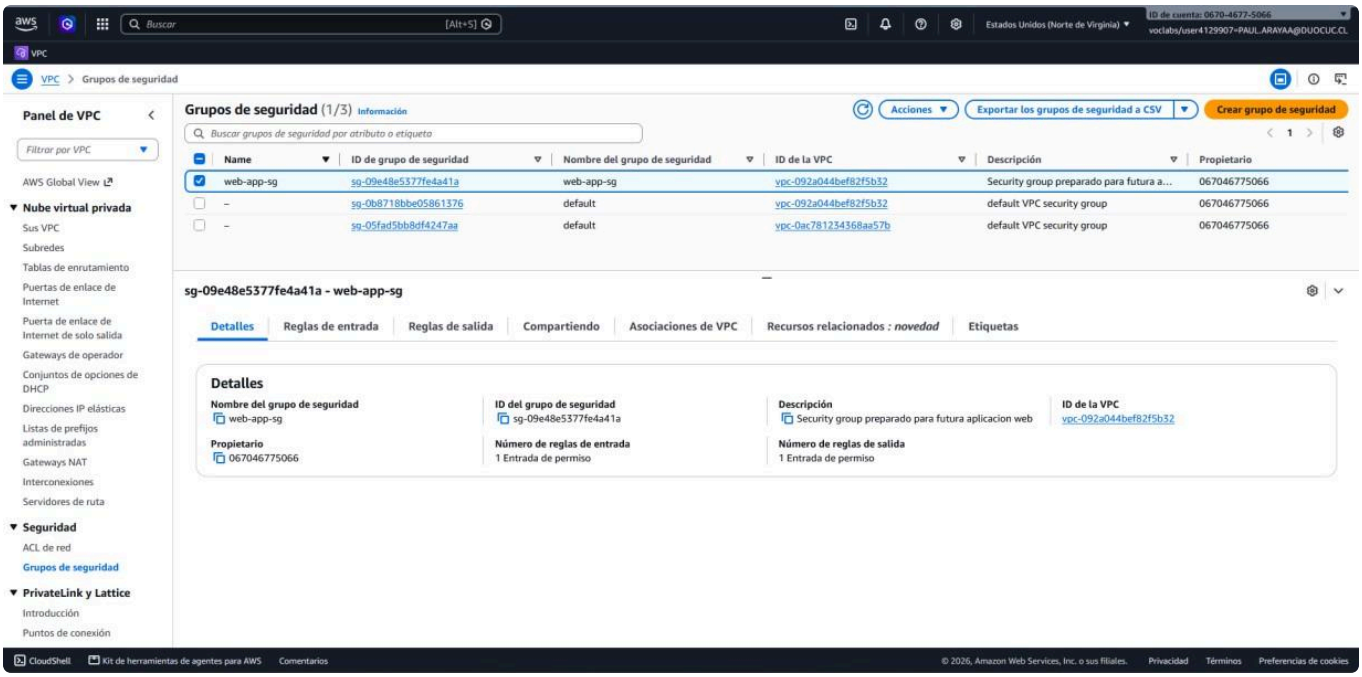

File Edit Selection View
parte2
EXPLORER
PARTE2
.terraform
.terraform.lock.hcl
main.tf
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
PS C:\Users\Duoc\Desktop\parte2> terraform plan
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create
Terraform will perform the following actions:
# aws_internet_gateway.igw will be created
+ resource "aws_internet_gateway" "igw" {
+ arn = (known after apply)
+ id = (known after apply)
+ owner_id = (known after apply)
+ region = "us-east-1"
+ tags = {
+ "Name" = "fastsolutions-igw"
}
+ tags_all = {
+ "Name" = "fastsolutions-igw"
}
+ vpc_id = (known after apply)
}
# aws_route_table.private_rt will be created
+ resource "aws_route_table" "private_rt" {
+ arn = (known after apply)
+ id = (known after apply)
+ owner_id = (known after apply)
+ propagating_vgws = (known after apply)
+ region = "us-east-1"
+ route = (known after apply)
+ tags = {
+ "Name" = "private-route-table"
}
+ tags_all = {
+ "Name" = "private-route-table"
}
+ vpc_id = (known after apply)
}
# aws_route_table.public_rt will be created
+ resource "aws_route_table" "public_rt" {
+ arn = (known after apply)
+ id = (known after apply)
+ owner_id = (known after apply)
+ propagating_vgws = (known after apply)
+ region = "us-east-1"
+ route = [
+ {
+ cidr_block = "0.0.0.0/0"
+ gateway_id = (known after apply)
# (12 unchanged attributes hidden)
},
]
+ tags = {
OUTLINE
TIMELINE
0 0 Live Share
Ln 117, Col 1 Spaces: 4 UTF-8 CRLF { } Plain Text Go Live

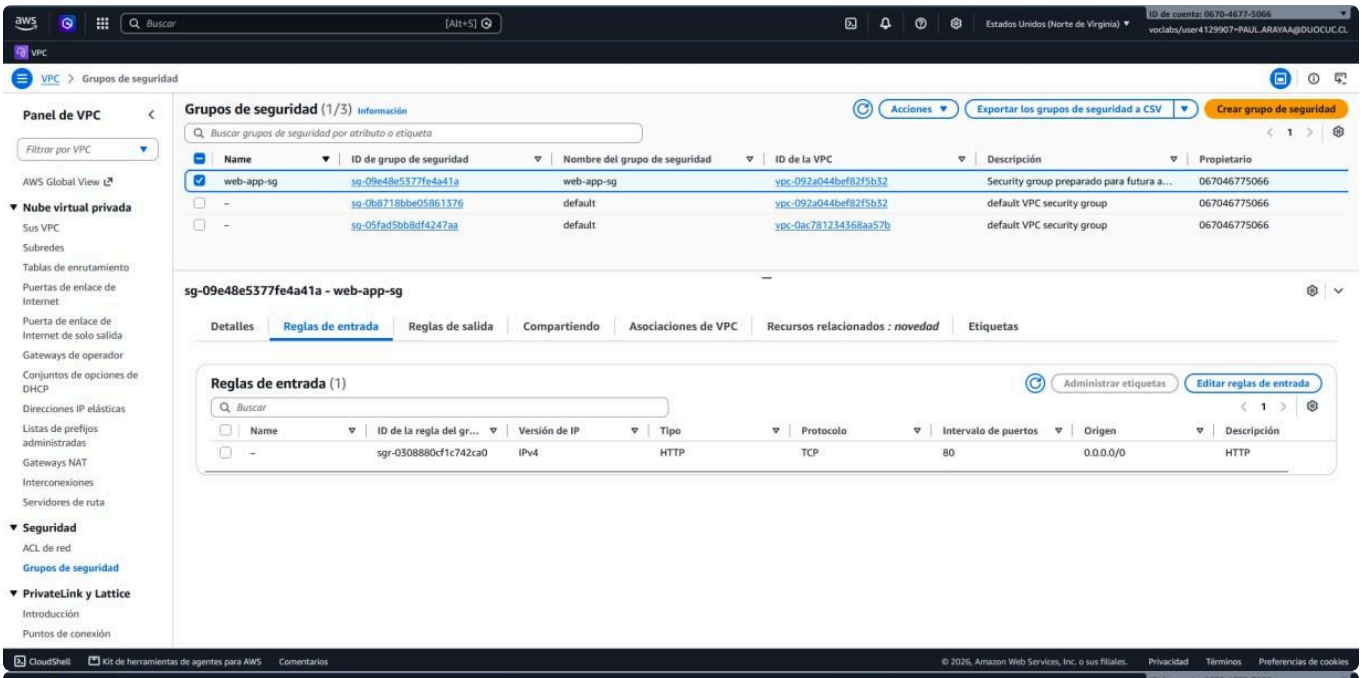

File Edit Selection View
parte2
EXPLORER
PARTE2
.terraform
.terraform.lock.hcl
main.tf
terraform.tfstate
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
+ enable_dns_support = true
+ enable_network_address_usage_metrics = (known after apply)
+ id = (known after apply)
+ instance_tenancy = "default"
+ ipv6_association_id = (known after apply)
+ ipv6_cidr_block = (known after apply)
+ ipv6_cidr_block_network_border_group = (known after apply)
+ main_route_table_id = (known after apply)
+ owner_id = (known after apply)
+ region = "us-east-1"
+ tags = {
+ "Name" = "fastsolutions-vpc"
}
+ tags_all = {
+ "Name" = "fastsolutions-vpc"
}
}
Plan: 13 to add, 0 to change, 0 to destroy.
Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.
Enter a value: yes
aws_vpc.main_vpc: Creating...
aws_vpc.main_vpc: Creation complete after 4s [id=vpc-092a044bef82f5b32]
aws_subnet.public_a: Creating...
aws_security_group.web_sg: Creating...
aws_subnet.public_b: Creating...
aws_internet_gateway.igw: Creating...
aws_route_table.private_rt: Creating...
aws_subnet.private_b: Creating...
aws_subnet.private_a: Creating...
aws_subnet.public_a: Creation complete after 1s [id=subnet-013be245056d32e96]
aws_subnet.public_b: Creation complete after 1s [id=subnet-05c824e13a4d09a75]
aws_subnet.private_a: Creation complete after 1s [id=subnet-062ac159f4db1bd91]
aws_subnet.private_b: Creation complete after 1s [id=subnet-0e66a4c28c275c2db]
aws_internet_gateway.igw: Creation complete after 1s [id=igw-0342739194ed2208c]
aws_route_table.private_rt: Creation complete after 1s [id=rtb-09275fec6cd141456]
aws_route_table.public_rt: Creating...
aws_route_table_association.priv_assoc_a: Creating...
aws_route_table_association.priv_assoc_b: Creating...
aws_route_table_association.priv_assoc_b: Creation complete after 1s [id=rtbassoc-0b8a4a671524a76e8]
aws_route_table_association.priv_assoc_a: Creation complete after 1s [id=rtbassoc-0880b09f6f16ad525]
aws_route_table.public_rt: Creation complete after 2s [id=rtb-04ff25d6b5a986f70]
aws_route_table_association.pub_assoc_b: Creating...
aws_route_table_association.pub_assoc_a: Creating...
aws_route_table_association.pub_assoc_b: Creation complete after 1s [id=rtbassoc-08840e688eec7eefe]
aws_route_table_association.pub_assoc_a: Creation complete after 1s [id=rtbassoc-0c8794b8fce773d3e]
aws_security_group.web_sg: Creation complete after 4s [id=sg-09e48e5377fe4a41a]
Apply complete! Resources: 13 added, 0 changed, 0 destroyed.
PS C:\Users\Duoc\Desktop\parte2>
OUTLINE
TIMELINE
0 0 Live Share
Ln 117, Col 1 Spaces: 4 UTF-8 CRLF { } Plain Text Go Live

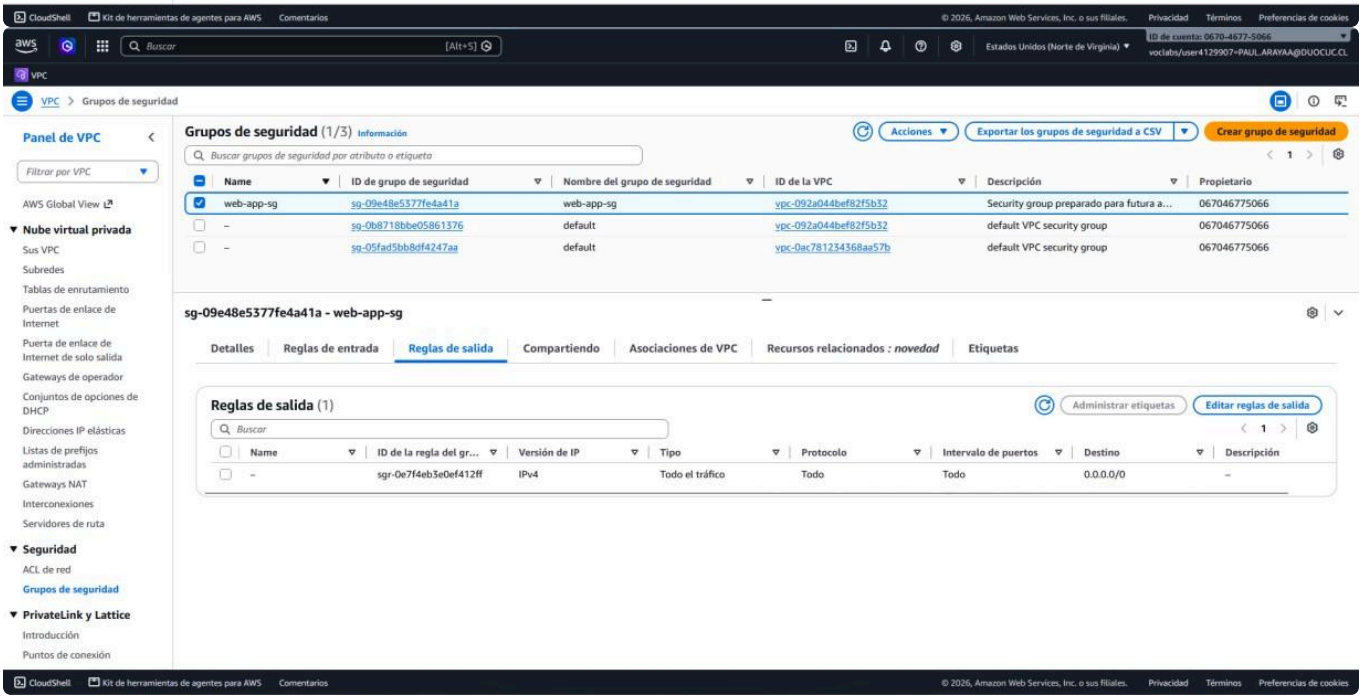
## Dashboard de AWS




## Security Group




Buscar
[Alt+S]
Estados Unidos (Norte de Virginia)
ID de cuenta: 0670-4677-5066
voclabs/user4129907=PAUL.ARAYAA@DUOCUC.CL
VPC
VPC > Grupos de seguridad
Panel de VPC
Filtrar por VPC
AWS Global View
Nube virtual privada
Sus VPC
Subredes
Tablas de enrutamiento
Puertas de enlace de Internet
Puerta de enlace de Internet de solo salida
Gateways de operador
Conjuntos de opciones de DHCP
Direcciones IP elásticas
Listas de prefijos administradas
Gateways NAT
Interconexiones
Servidores de ruta
Seguridad
ACL de red
Grupos de seguridad
PrivateLink y Lattice
Introducción
Puntos de conexión
Grupos de seguridad (1/3) Información
Acciones
Exportar los grupos de seguridad a CSV
Crear grupo de seguridad
Buscar grupos de seguridad por atributo o etiqueta
Name
ID de grupo de seguridad
Nombre del grupo de seguridad
ID de la VPC
Descripción
Propietario
web-app-sg
sg-09e48e5377fe4a41a
web-app-sg
vpc-092a044bef82f5b32
Security group preparado para futura a...
067046775066
sg-0b8718bbe05861376
default
vpc-092a044bef82f5b32
default VPC security group
067046775066
sg-05fad5bb8df4247aa
default
vpc-0ac781234368aa57b
default VPC security group
067046775066
sg-09e48e5377fe4a41a - web-app-sg
Detalles
Reglas de entrada
Reglas de salida
Compartiendo
Asociaciones de VPC
Recursos relacionados : novedad
Etiquetas
Reglas de entrada (1)
Administrar etiquetas
Editar reglas de entrada
Buscar
Name
ID de la regla del gr...
Versión de IP
Tipo
Protocolo
Intervalo de puertos
Origen
Descripción
sgr-0308880cf1c742ca0
IPv4
HTTP
TCP
80
0.0.0.0/0
HTTP
CloudShell
Kit de herramientas de agentes para AWS
Comentarios
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Preferencias de cookies


Buscar
[Alt+S]
Estados Unidos (Norte de Virginia)
ID de cuenta: 0670-4677-5066
voclabs/user4129907=PAUL.ARAYAA@DUOCUC.CL
VPC
VPC > Grupos de seguridad
Panel de VPC
Filtrar por VPC
AWS Global View
Nube virtual privada
Sus VPC
Subredes
Tablas de enrutamiento
Puertas de enlace de Internet
Puerta de enlace de Internet de solo salida
Gateways de operador
Conjuntos de opciones de DHCP
Direcciones IP elásticas
Listas de prefijos administradas
Gateways NAT
Interconexiones
Servidores de ruta
Seguridad
ACL de red
Grupos de seguridad
PrivateLink y Lattice
Introducción
Puntos de conexión
Grupos de seguridad (1/3) Información
Acciones
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Crear grupo de seguridad
Buscar grupos de seguridad por atributo o etiqueta
Name
ID de grupo de seguridad
Nombre del grupo de seguridad
ID de la VPC
Descripción
Propietario
web-app-sg
sg-09e48e5377fe4a41a
web-app-sg
vpc-092a044bef82f5b32
Security group preparado para futura a...
067046775066
sg-0b8718bbe05861376
default
vpc-092a044bef82f5b32
default VPC security group
067046775066
sg-05fad5bb8df4247aa
default
vpc-0ac781234368aa57b
default VPC security group
067046775066
sg-09e48e5377fe4a41a - web-app-sg
Detalles
Reglas de entrada
Reglas de salida
Compartiendo
Asociaciones de VPC
Recursos relacionados : novedad
Etiquetas
Reglas de salida (1)
Administrar etiquetas
Editar reglas de salida
Buscar
Name
ID de la regla del gr...
Versión de IP
Tipo
Protocolo
Intervalo de puertos
Destino
Descripción
sgr-0e7f4eb3e0ef412ff
IPv4
Todo el tráfico
Todo
Todo
0.0.0.0/0
–
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Kit de herramientas de agentes para AWS
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Terraform Destroy

